

Climate fluctuations shape primary-school academic performance through differential pathways: evidence from Côte d'Ivoire

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Abstract

Climate change is increasingly framed as a threat to human capital, yet evidence on how normalised, non-disaster climate fluctuations shape day-to-day educational quality in low-income African settings remains sparse. Here we combine 67,135 standardised dictation observations gathered monthly across primary schools in Côte d’Ivoire (October 2022–June 2025) with sub-national rainfall indicators derived from CHIRPS, and use a large-language-model (LLM) pipeline to anchor variable choices in policy and scientific evidence. We then estimate causal effects with a ladder of identification strategies—two-way fixed effects, Heckman sample-selection correction, linear double/debiased machine learning and a causal forest. Rainfall affects academic performance through three differential pathways: short-term precipitation depresses performance (rainy-day effect), long-term precipitation enhances it (income effect), and inter-monthly variability erodes it (uncertainty shock). Household economic conditions—proxied by school type and location—both directly determine performance and moderate its sensitivity to climate. The causal forest reveals that public–rural pupils are most exposed in pass-rate terms, whereas private-school pupils are most exposed in excellence-rate terms. These results support targeted, climate-resilient education policy aligned with Sustainable Development Goals 4 and 13.

Climate change is reshaping the conditions under which human capital accumulates^{1–3}. The 2025 United Nations Sustainable Development Goals Report indicates that, as of 2023, 272 million children and adolescents worldwide remained out of school, and that in 2019 only 58% of children achieved minimum reading proficiency. Learning systems in low-income countries are also disproportionately exposed to climate shocks^{4,5}. Most existing evidence has focused on the *extensive* margin of education—enrolment, dropout, attainment^{6–8}—and on rare, headline-grabbing disasters such as floods, droughts and tropical cyclones^{9–13}. By contrast, evidence on how *normalised* climate fluctuations—routine deviations of temperature, rainfall and variability from local norms—affect the *intensive* margin of learning is comparatively scarce, especially in sub-Saharan Africa, which receives only a small fraction of global climate-research funding^{14–17}.

A second gap concerns identification. Climate-education studies face well-known endogeneity threats from unobserved regional heterogeneity, time-varying macroeconomic, political, and educational-policy conditions, selective participation, and high-dimensional confounding by household and school characteristics^{18–22}. A third gap concerns measurement: in linguistic, institutional and policy contexts as heterogeneous as Côte d’Ivoire, the choice of which climate indicator to use and which household variables to proxy is itself a knowledge-intensive task, traditionally handled by manual literature review.

We address all three gaps in a single study. We use 67,135 observations of standardised French dictation results, collected monthly across Côte d’Ivoire’s primary schools by UNICEF Côte d’Ivoire and the Ministry of National Education and Literacy in 2022–2025, as a fine-grained intensive-margin measure of learning. We pair these with CHIRPS-derived rainfall indicators released by the United Nations Office for the Coordination of Humanitarian Affairs (OCHA) at the sub-national level, and decompose precipitation into three complementary dimensions: short-term level (one-month accumulation), long-term level (three-month accumulation) and inter-monthly variability^{23,24}. To eliminate the bottleneck of manual variable identification, we deploy an LLM (DeepSeek-V3.2) within a chain-of-thought pipeline that decomposes evidence mining into a fact-gathering phase and an expert-reasoning phase^{25–31}.

For identification, we combine two-way (administrative-district by date) fixed effects with a Heckman selection correction, a linear double/debiased machine-learning (LinearDML) estimator that orthogonalises high-dimensional confounders^{21,32,33}, and a causal forest that recovers heterogeneous treatment effects without imposing a parametric interaction^{22,34,35}. Figure 1 summarises the pipeline.

Three findings emerge. First, rainfall affects academic performance through *three differential pathways*: short-term precipitation reduces both the pass and the excellence rate (rainy-day effect), long-term precipitation raises both (income effect), and inter-monthly variability lowers both (uncertainty shock). Second, household economic conditions, proxied by whether the school is private or public and urban or rural, both directly determine performance and moderate its sensitivity to

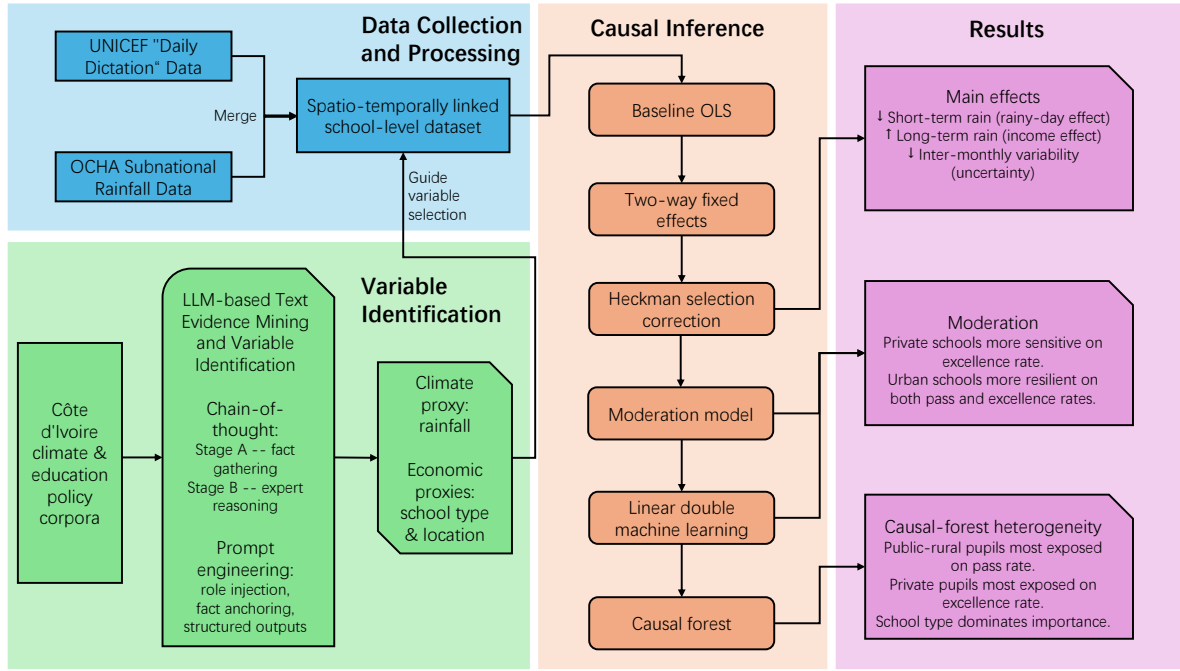


Figure 1. Multi-source pipeline for identifying causal effects of climate fluctuations on academic performance in Côte d’Ivoire. The framework proceeds in four stages. First, UNICEF Daily Dictation records are merged with OCHA sub-national rainfall data to construct a spatio-temporally linked school-level dataset. Second, Côte d’Ivoire climate and education policy corpora are processed through an LLM-based text evidence-mining and variable-identification module, in which chain-of-thought prompting separates fact gathering from expert reasoning and uses prompt-engineering safeguards including role injection, fact anchoring and structured outputs. This stage guides the selection of rainfall as the climate proxy and school type and location as economic proxies. Third, causal inference is implemented through a progressive sequence of estimators, moving from baseline OLS to two-way fixed effects, Heckman selection correction, a moderation model, LinearDML and a causal forest. Fourth, the resulting estimates are organised into three classes of findings: main rainfall effects, moderation by household economic conditions and causal-forest heterogeneity.

climate. Third, the causal forest reveals a sharply heterogeneous structure: public-rural pupils are the most exposed on the pass rate, whereas private pupils—who on average outperform their peers—are the most exposed on the excellence rate. School type is the single largest source of heterogeneity, especially for the interaction between excellence rate and inter-monthly variability.

Results

Setting and dataset

Côte d’Ivoire spans two climate regimes. Coastal and southern districts are dominated by the seasonal migration of the Intertropical Convergence Zone (ITCZ), producing two rainy seasons (April–July and October–November); northern districts experience a single, shorter rainy season under a tropical savannah regime^{36–40}. Consistent with this climate structure, the LLM-based evidence-mining pipeline isolated rainfall as the dominant climate indicator in Côte d’Ivoire and identified school type (private vs. public) and school location (urban vs. rural) as the most credible proxies for household economic conditions^{41,42}.

The Daily Dictation programme records, for every grade–round combination in a participating school, the number of pupils who score 0, 1–5, or ≥ 6 errors, broken down by gender⁴³. From these we compute a *pass rate* (0–5 errors) and an *excellence rate* (0 errors). After deduplication, language and grade matching, and reshaping by gender, the dataset contains 67,135 observations across 24 Regions, 34 Departments and three school years; Extended Data Fig. ED1 reports the year and grade composition.

Rainfall data come from the CHIRPS-based OCHA dataset, which provides ten-day-step accumulations and historical anomalies at the Department level (1981–2025)^{23,24}. We standardise three indicators: *Normr1h* for the normalised one-month accumulation, *Normr3h* for the normalised three-month accumulation, and *NormFluctMonth* for the normalised month-to-month change. The first proxies short-term “rainy-day” shocks, the second captures longer-run precipitation conditions, and the third proxies the unpredictability of the rainfall regime^{3,6,44}. Extended Data Tables ED1–ED3 report the descriptive statistics, correlation matrix and variance-inflation factors (all VIFs < 2).

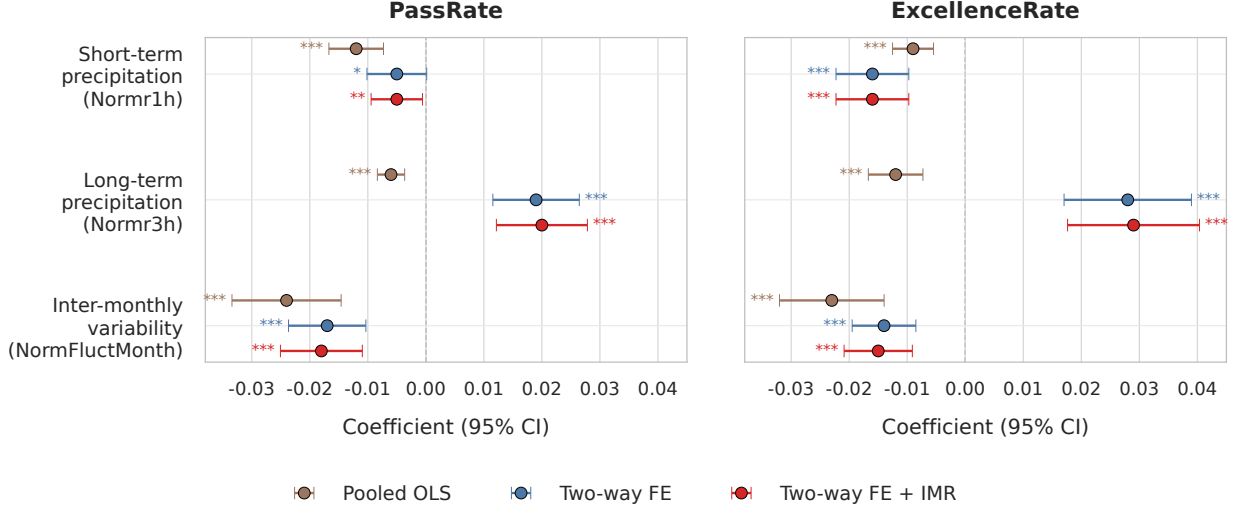
LLM-based variable identification

A non-trivial methodological contribution is that the choice of climate indicator and of household-economy proxies was made by a large-language model anchored in source material. We curated 24 climate documents (Wikipedia, Britannica, PIK reports, peer-reviewed monsoon studies) and 27 education documents (UNESCO, UNICEF and MENA reports) and passed them in two chains-of-thought through DeepSeek-V3.2 (temperature=0.2). In Stage A the model extracted, in the original language, sentences that addressed predefined topics (temperature, rainfall, winds, ITCZ migration; private vs. public schooling, urban–rural gaps). In Stage B the model assumed the role of a senior domain expert, performed cross-document triangulation and produced a written rationale anchored to the Stage A excerpts^{25–27,29}.

This two-stage chain-of-thought, combined with role injection, fact anchoring and structured output, scientifically isolated rainfall as the binding climate dimension in Côte d’Ivoire (temperature, wind speed and pressure showed limited intra-annual variation) and household economy as best proxied by school type and location. Human reviewer independently verified all cited excerpts and the reasoning trace. The full workflow diagram, prompts, retrieved excerpts and machine reasoning are provided in Supplementary Section S1.

Main causal effects on academic performance

Climate–academic performance estimates change substantially as the empirical specification is progressively strengthened (Fig. 2). The pooled OLS models provide a descriptive baseline, but they do not address key sources of endogeneity in this setting, including unobserved regional heterogeneity, common time-varying shocks and non-random participation in the Daily Dictation programme. In these baseline models, all three rainfall indicators are negatively associated with both outcomes. After introducing two-way (District $\times$ Date) fixed effects to absorb time-invariant district characteristics and date-specific macro trends, the coefficient on *Normr3h* turns positive.



Note: horizontal lines show 95% confidence intervals; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 2. Main effects of standardised precipitation on academic performance change across identification strategies. Point estimates with 95% confidence intervals for three standardised rainfall indicators (short-term level Normr1h, long-term level Normr3h, inter-monthly variability NormFluctMonth) on the pass rate (left) and the excellence rate (right), under pooled OLS (brown), two-way District $\times$ Date fixed effects (blue), and the preferred two-way FE + Inverse Mills Ratio specification (red). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Further adding the Inverse Mills Ratio from the Heckman selection model to correct for selective participation and successful reporting leaves the qualitative pattern unchanged and slightly sharpens the estimates. The Inverse Mills Ratio itself is significantly negative on both outcomes, confirming that schools selected into the programme—and that report successfully—are systematically those with better infrastructure and resources (Extended Data Table ED4).

In the preferred specification (two-way fixed effects with Heckman correction; Extended Data Table ED5), Normr1h is significantly negative on both outcomes ($\hat{\beta}_{\text{pass}} = -0.005$, $p < 0.05$; $\hat{\beta}_{\text{exc}} = -0.016$, $p < 0.01$); Normr3h is significantly positive ($\hat{\beta}_{\text{pass}} = 0.020$, $p < 0.01$; $\hat{\beta}_{\text{exc}} = 0.029$, $p < 0.01$); and NormFluctMonth is significantly negative ($\hat{\beta}_{\text{pass}} = -0.018$, $p < 0.01$; $\hat{\beta}_{\text{exc}} = -0.015$, $p < 0.01$). These sign changes relative to the pooled OLS baseline suggest that the simple cross-sectional association is confounded and should not be interpreted causally. Once regional heterogeneity, date-specific shocks and sample-selection bias are accounted for, the estimates reveal three distinct rainfall–learning relationships: short-term precipitation is associated with lower performance, longer-run precipitation is associated with higher performance, and inter-monthly variability is associated with lower performance. This pattern is consistent with short-term rainy-day disruption, longer-run rainfall conditions that may improve agricultural and household-resource environments, and uncertainty shocks that interfere with school scheduling and learning continuity.

Female students score higher than male students on both rates ($\hat{\beta}_{\text{pass}} = -0.011$, $\hat{\beta}_{\text{exc}} = -0.009$, both $p < 0.01$), echoing well-documented gender gaps in language-intensive tasks^{45–48}, and pass rates rise systematically with grade level, mirroring developmental and accumulative learning effects^{49–51}.

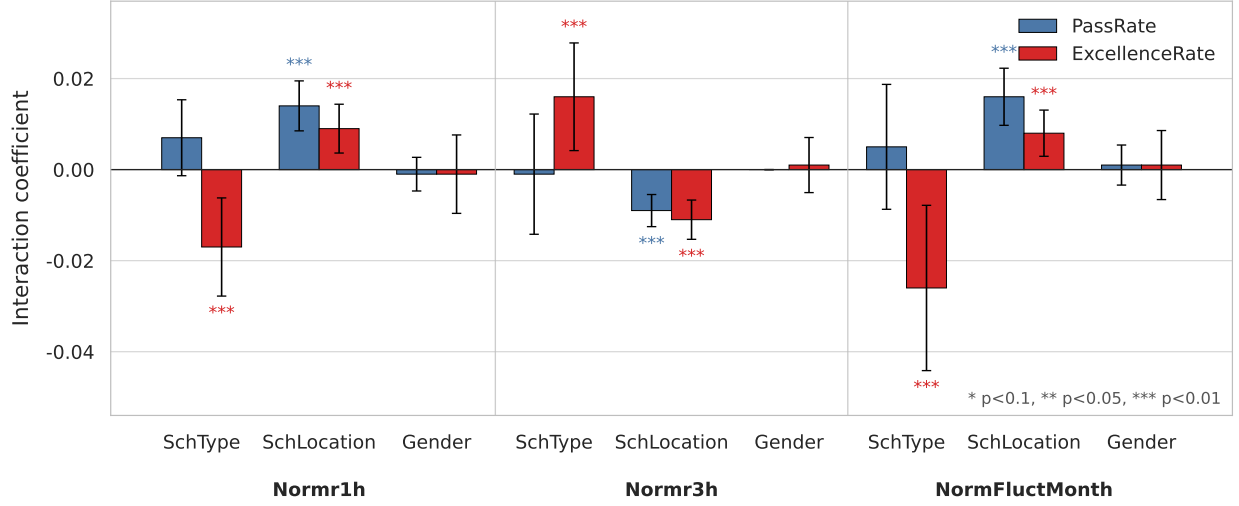


Figure 3. Moderation of standardised rainfall effects by school characteristics and student gender. Interaction coefficients from the moderation regression. Within each rainfall-indicator group, bars show interactions with school type (SchType: private vs. public, private=1), school location (SchLocation: urban vs. rural, urban=1), and student gender (Gender: male vs. female, male=1); error bars represent 95% confidence intervals. Stars denote significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Direct and moderating effects of household economic conditions on academic performance

Private-school pupils score higher on both outcomes (pass-rate gain of 0.064, excellence-rate gain of 0.040, both $p < 0.01$), while urban-school pupils score higher on the excellence rate (gain of 0.019, $p < 0.01$) but slightly lower on the pass rate (loss of 0.009, $p < 0.01$) (Extended Data Table ED6). The moderation analysis (Fig. 3, Extended Data Table ED6) shows that, although gender has no detectable moderating role, both school-type and location-type interact significantly with rainfall:

(i) Private-school pupils are more sensitive on the excellence rate, with significantly negative interactions for short-term level (-0.017 , $p < 0.01$) and inter-monthly variability (-0.026 , $p < 0.01$), and a significantly positive interaction for long-term level ($+0.016$, $p < 0.01$). Their competitive advantage at the top of the distribution thus narrows under climate stress.

(ii) Urban-school pupils are consistently buffered from climate fluctuations: for all three rainfall indicators, the interaction terms run opposite to the corresponding main effects. Interactions with Normr1h and NormFluctMonth are positive on both the pass rate ($+0.014$ and $+0.016$, both $p < 0.01$) and the excellence rate ($+0.009$ and $+0.008$, both $p < 0.01$), offsetting short-term and variability-related losses. The interaction with Normr3h is negative on both outcomes (-0.009 and -0.011 , both $p < 0.01$), implying weaker dependence on favourable longer-run rainfall. Urban schools therefore display lower overall climate sensitivity and stronger resilience.

These results are quantitatively very similar in the LinearDML specification, which uses LassoCV to residualise both the outcome and each rainfall treatment with respect to high-dimensional controls, including the Inverse Mills Ratio, grade dummies, district dummies and school-year/date controls, before estimating the interaction effects (Extended Data Table ED7). Out of the nine interaction

terms, eight retain the same sign under the LinearDML specification, while the remaining one becomes marginally non-significant but remains directionally consistent.

Heterogeneous effects from a causal forest

To probe heterogeneity beyond the imposed linear interactions, we estimated six causal forests (two outcomes $\times$ three treatments) with 2,000 trees, a minimum leaf size of 100, a minimum within-leaf treatment-variance fraction of 0.1 and three-fold cross-fitting. The standardised rainfall variables are continuous treatments; school type, school location and student gender define the effect-modifying vector X ; and the Inverse Mills Ratio, the other two rainfall indicators, grade dummies, district dummies and date dummies form the orthogonalising control vector W ^{34,52,53}. Figure 4 (and Fig. ED2 for the frequency distribution of conditional average treatment effects) reports the group-level conditional average treatment effects (CATEs) for the eight SchType $\times$ SchLocation $\times$ Gender combinations.

Three patterns are striking. First, the negative pass-rate effect of short-term rainfall is concentrated in public-rural pupils ($CATE = -0.015$ for girls and -0.019 for boys, both $p < 0.01$). Public-urban pupils show no significant exposure. Private pupils display positive point estimates; the estimate is statistically significant only for private-urban boys, while the other private-school estimates remain imprecise because of small subgroup shares. The income-channel positive effect of long-term rainfall is similarly concentrated in public-school groups (CATEs of about $+0.015$ to $+0.020$), again consistent with a household-income mechanism rooted in subsistence agriculture^{6,44}.

Second, the negative excellence-rate effect is largest, in absolute size, among private-school pupils. For inter-monthly variability the CATE reaches -0.074 for private-urban girls, -0.056 for private-urban boys and -0.053 for private-rural girls. The same private-school dominance shows up for the rainy-day effect (-0.041 , -0.035 , -0.035 , respectively). Long-term rainfall, by contrast, is associated with the largest *positive* CATEs in private-school groups ($+0.060$ for private-urban girls), suggesting that high-baseline private-school pupils benefit most when prolonged rainfall improves agricultural income channels in their broader community.

Third, the feature-importance heatmap (Fig. 5) attributes the bulk of the heterogeneity to school type. School type is the dominant moderator in every model and is overwhelmingly so for the ExcellenceRate $\sim$ NormFluctMonth model (importance 0.780). This is an independent confirmation of the moderation regressions: household economic conditions, not gender, drive the heterogeneous climate response.

Robustness

The qualitative pattern of three differential pathways and household-economic moderation survives every robustness check we performed. Winsorising the three rainfall indicators at the 1% and 5% level changes neither the sign nor the significance of the main effects in either the moderation or the LinearDML specifications (Supplementary Tables S1–S2). Restricting to an effective sample of school-round observations with more than 10 or 15 dictation participants leaves the picture intact (Supplementary Tables S3–S4). Replacing District fixed effects with finer-grained Region

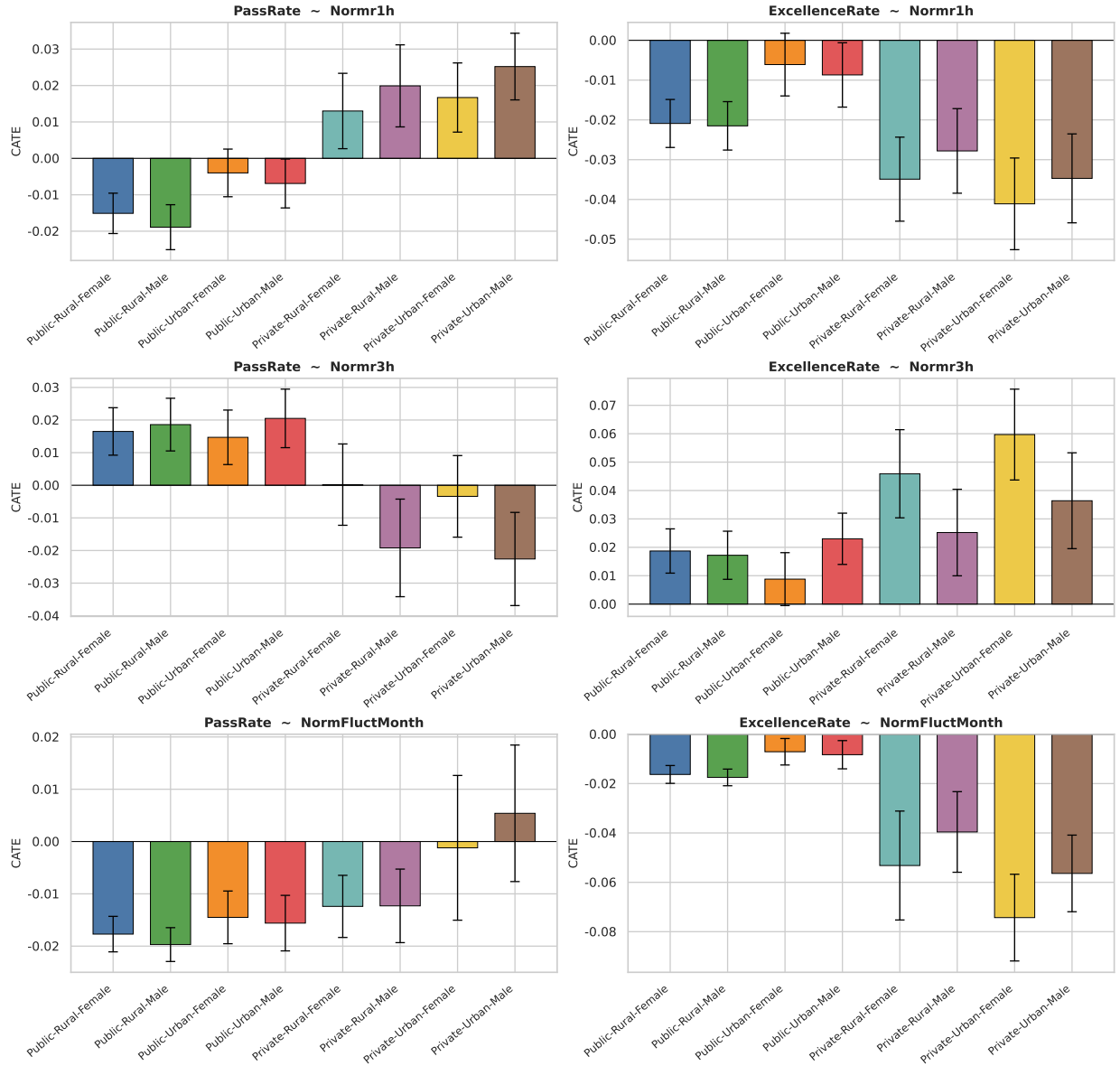


Figure 4. Group-level conditional average treatment effects (CATEs) of standardised precipitation on academic performance. Each panel shows CATEs estimated from one of six causal-forest models (rows: short-term level Normr1h, long-term level Normr3h, inter-monthly variability NormFluctMonth; columns: pass rate, excellence rate) across the eight subgroups defined by school type, school location and student gender. Error bars are point-wise standard errors derived from the honest causal forest. Public-rural pupils carry most of the negative pass-rate effect of short-term rain, while private-school pupils carry most of the negative excellence-rate effect of variability and short-term rain.

fixed effects yields nearly identical coefficients (Supplementary Table S5). Re-estimating the causal forests on the winsorised and effective-sample data preserves the public-rural-vs-private gradient documented in the preferred per-group CATEs (Supplementary Table S6).

Discussion

Côte d’Ivoire provides a clean natural laboratory for studying the intensive margin of learning under climate stress because rainfall, rather than temperature, is the binding climate dimension, and

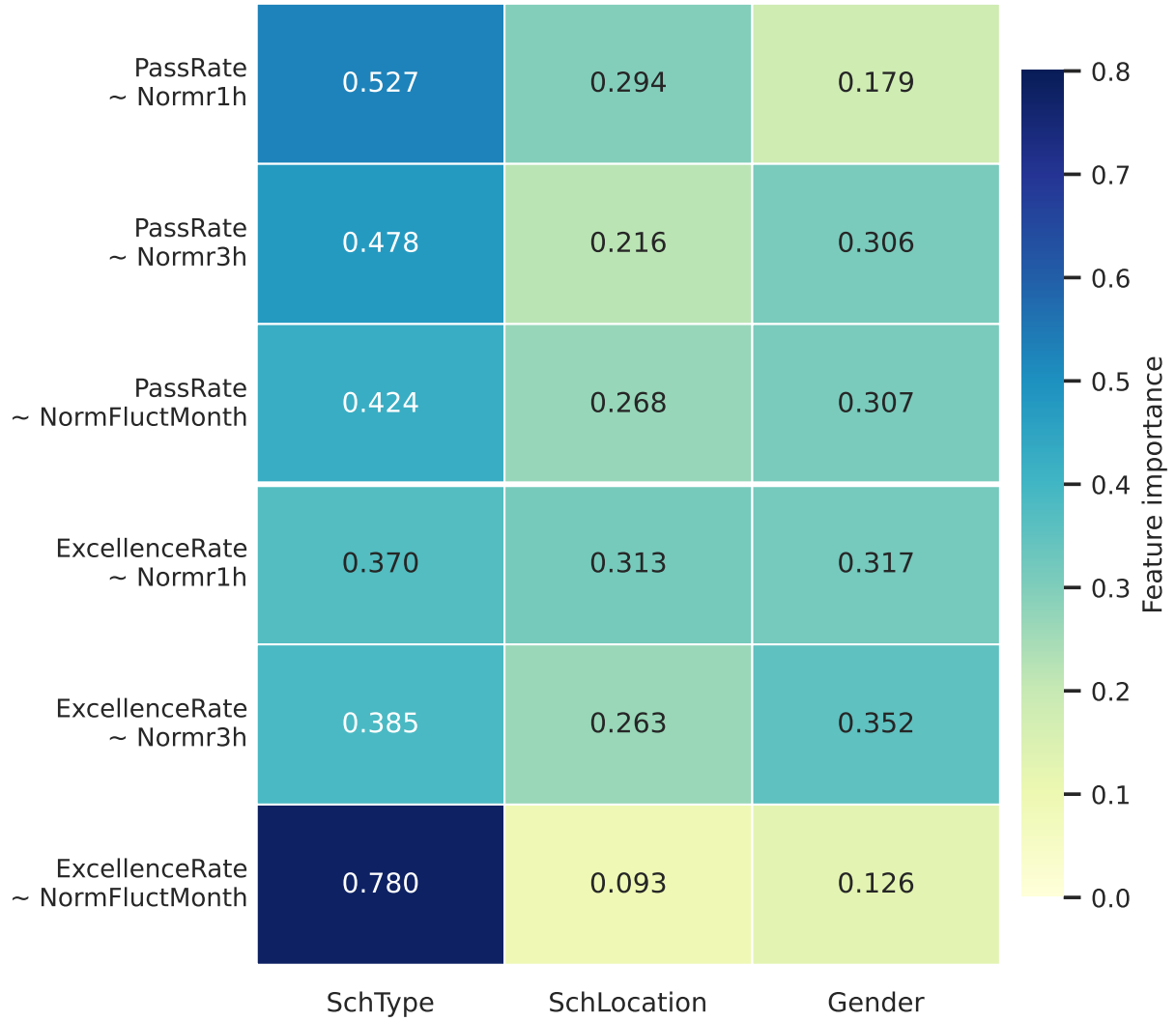


Figure 5. Feature importance from the six causal forests confirms that school type drives most of the treatment-effect heterogeneity. Cell entries are normalised feature-importance scores for school type, school location and student gender across the six outcome $\times$ treatment combinations. School type is the dominant moderator throughout, and is overwhelmingly so for the excellence-rate response to inter-monthly variability (importance 0.780).

because UNICEF’s monthly Daily Dictation system provides standardised, high-frequency outcomes that are otherwise rare in sub-Saharan Africa^{12,41}. Three substantive contributions follow.

First, our results document that, beyond the well-studied extensive-margin effects of disasters^{9–12}, normalised climate variability has measurable, signed, mechanism-specific impacts on day-to-day learning quality. The differential signatures we identify—negative for short-term rainfall (rainy-day disruption channel), positive for long-term rainfall (household-income channel), negative for inter-monthly variability (uncertainty channel)—correspond to three distinct economic and psychological mechanisms documented separately in the literature^{6,15,44,54–57}. The fact that they coexist in the same dataset, with different relative weights on the pass and excellence rates, suggests that policy responses cannot be one-dimensional.

Second, the moderation and causal-forest analyses together show that the policy-relevant unit of heterogeneity is the household–school joint distribution, not individual demographics. Gender does not moderate; school type and location do, and they do so in opposite directions on the pass and excellence rates. Public–rural pupils—who already trail in basic competence—are most at risk of losing the bottom margin when short-term and variable rainfall increase. Private-school pupils—who lead at the top—are most at risk of losing the top margin when the regime becomes erratic. Climate-resilient education policy must therefore target two distinct ends of the achievement distribution with different instruments.

Third, the LLM-anchored variable-identification pipeline is itself a methodological contribution. In contexts where the relevant literature is dispersed across languages (French, English, Chinese, technical documents and websites) and across academic, intergovernmental and government sources, chain-of-thought prompting with role injection and fact anchoring can recover scientifically defensible variable choices in hours rather than weeks, while preserving full traceability for human verification^{27–30}.

Operationally, our findings translate into a five-step “climate-risk identification — early warning — teaching schedule — resource compensation — impact assessment” policy loop that the Ministry of National Education and Literacy could adopt at low marginal cost. When standardised rainfall exceeds one standard deviation, schools with low public infrastructure or high social-economic vulnerability should pre-emptively reschedule, deploy off-line teaching materials and assign mobile teacher support, while monitoring rebound by integrating outcome reporting with the rainfall trigger. This approach embeds climate adaptation into the routine cadence of school administration and is directly translatable to other sub-Saharan settings facing similar rainfall regimes—and to climate-vulnerable regions of middle-income countries.

The study has limitations. Our climate exposure is captured at ten-day granularity; richer daily or hourly station data could separate brief intense storms from steady drizzle that yield identical ten-day totals. Our 67,135 observations span three school years but average only 2.6 rounds per school, precluding genuine panel-data treatments and ruling out the estimation of state-dependence or hysteresis^{18,19}. Finally, although the causal forest captures non-parametric heterogeneity in continuous treatment effects, threshold or non-monotonic responses (for example, to extreme rainfall) would be better explored with semiparametric dose–response models or deep-learning architectures^{58,59}. We see these as natural extensions, especially as longer panels of the Daily Dictation system accumulate.

Methods

Data sources and processing

Educational outcomes. The Daily Dictation programme (*dictée du jour*) is run jointly by UNICEF Côte d’Ivoire and the country’s Ministry of National Education and Literacy (MENA)⁴³. Every participating primary school administers standardised French dictation tests at monthly

frequency, then uploads, for each grade-round combination, the count of pupils with 0, 1–5 and ≥ 6 errors, broken down by gender. We obtained the full upload table for October 2022–June 2025 through an administrator account with formal authorisation from the UNICEF Côte d’Ivoire Country Representative. The raw Daily Dictation extract contains 67,135 school-round-gender-grade observations. After binary-encoding categorical variables, reshaping by gender, harmonising field names across years, deduplicating fuzzy school-name matches and removing missing or implausible records after rainfall and IMR matching, the main analytical regressions use 64,743 complete observations. The Heckman selection frame contains 19,495 schools, 34 Departments, 24 Regions and 14 Districts. The pass rate is the share of pupils with 0–5 errors and the excellence rate is the share of pupils with 0 errors:

$$\text{PassRate} = \frac{n_{0 \text{ err}} + n_{1-5 \text{ err}}}{N}, \quad \text{ExcellenceRate} = \frac{n_{0 \text{ err}}}{N}. \quad (1)$$

Climate indicators. Sub-national rainfall comes from the OCHA HDX dataset *Côte d’Ivoire – Rainfall Indicators at Subnational Level*, derived from CHIRPS satellite and station data and aggregated to Department-level ten-day accumulations from 1981 to 2025^{23,24}. We construct three standardised indicators:

$$\begin{aligned} \text{Normr1h}_{i,t} &= \frac{r1h_{i,t} - \mu_{r1h}}{\sigma_{r1h}}, & \text{Normr3h}_{i,t} &= \frac{r3h_{i,t} - \mu_{r3h}}{\sigma_{r3h}}, \\ \text{FluctMonth}_{i,t} &= \frac{|r1h_{i,t} - r1h_{i,t-3}|}{r1h_{i,t}}, & \text{NormFluctMonth}_{i,t} &= \frac{\text{FluctMonth}_{i,t} - \mu_{\text{FluctMonth}}}{\sigma_{\text{FluctMonth}}}. \end{aligned} \quad (2)$$

where $r1h$ ($r3h$) denotes one-month (three-month) cumulative rainfall at the school-round. Because the rainfall series is released at ten-day frequency, $t - 3$ denotes the previous month in the construction of *FluctMonth*. We link rainfall and education records by the Department code provided in the HDX administrative boundaries⁶⁰. Definitions of the variables and model symbols used throughout the manuscript are collected in Supplementary Section S3.

Large-language-model variable identification

To anchor variable choices in source material, we curated 24 climate documents (PIK climate-risk profile, Britannica, Wikipedia, peer-reviewed monsoon papers^{36–39}) and 28 education documents (UNESCO, UNICEF, MENA, and Chinese-language briefs^{41–43,61}), digitised them with *ocrmypdf*, chunked them to fit the LLM context window and passed them in two stages through DeepSeek-V3.2 (temperature = 0.2)²⁵.

Stage A used a fact-gathering system prompt for each domain and asked the model to return, as a Python list, sentences in the original language that addressed the predefined topics; this stage was instructed never to translate or invent text^{26,27,29}. Stage B passed all retrieved excerpts to the model as a user prompt while injecting an “expert in climate science” or “expert in education economics” role; the model was required to cross-validate the excerpts, derive a final answer and explicitly cite the supporting documents and quotations^{28,30}. Human reviewers re-read each cited

excerpt in its original document; in no case was the model found to fabricate evidence. The full Stage A and Stage B outputs are provided as a Supplementary appendix.

Models

Baseline OLS. Let i index a school-round-gender-grade observation and t the date. The baseline pooled-OLS model is

$$\begin{aligned} Y_{i,t} = & \beta_0 + \beta_1 \text{Normr1h}_{i,t} + \beta_2 \text{Normr3h}_{i,t} + \beta_3 \text{NormFluctMonth}_{i,t} \\ & + \beta_4 \text{SchType}_i + \beta_5 \text{SchLocation}_i + \beta_6 \text{Gender}_i \\ & + \sum_{g=1}^5 \rho_g G_{i,g} + \varepsilon_{i,t}, \end{aligned} \quad (3)$$

where $Y_{i,t}$ is PassRate or ExcellenceRate and $G_{i,g}$ are five grade dummies for the six grade levels.

Two-way fixed effects. We augment equation (3) with District (DistrictFE) and Date (DateFE) fixed effects to absorb time-invariant regional heterogeneity and macro-temporal trends:

$$\begin{aligned} Y_{i,t} = & \beta_0 + \beta_1 \text{Normr1h}_{i,t} + \beta_2 \text{Normr3h}_{i,t} + \beta_3 \text{NormFluctMonth}_{i,t} \\ & + \beta_4 \text{SchType}_i + \beta_5 \text{SchLocation}_i + \beta_6 \text{Gender}_i \\ & + \sum_{g=1}^5 \rho_g G_{i,g} + \text{DistrictFE}_i + \text{DateFE}_t + \varepsilon_{i,t}. \end{aligned} \quad (4)$$

Heckman selection correction. Participation in the Daily Dictation programme and the reliability of upload are both plausibly non-random, so we apply a Heckman two-stage correction⁶². The first stage estimates a probit for participation conditional on school infrastructure (electricity, multimedia room, internet connection) and local project-participation density, exclusion restrictions that affect participation but, conditional on observables, not the outcome:

$$\begin{aligned} D_i^* = & \gamma_0 + \gamma_1 \text{SchType}_i + \gamma_2 \text{SchLocation}_i \\ & + \gamma_3 \text{ElectrifiedSch}_i + \gamma_4 \text{MultimediaRoom}_i \\ & + \gamma_5 \text{InternetSch}_i + \gamma_6 \text{ProjectDensity}_i + \nu_i, \\ D_i = & \mathbf{1}\{D_i^* > 0\}. \end{aligned} \quad (5)$$

Equivalently, for

$$z_i = (1, \text{SchType}_i, \text{SchLocation}_i, \text{ElectrifiedSch}_i, \text{MultimediaRoom}_i, \text{InternetSch}_i, \text{ProjectDensity}_i), \quad (6)$$

we define

$$\begin{aligned} \Pr(D_i = 1 \mid z_i) &= \Pr(\nu_i > -z_i' \gamma) = \Phi(z_i' \gamma), \\ \text{IMR}_i &= \frac{\phi(z_i' \hat{\gamma})}{\Phi(z_i' \hat{\gamma})}. \end{aligned} \quad (7)$$

where D_i indicates that school i both participated and reported, and ϕ and Φ are the standard-normal density and cumulative distribution functions. We then add IMR_i to equation (4):

$$\begin{aligned} Y_{i,t} = & \beta_0 + \beta_1 \text{Normr1h}_{i,t} + \beta_2 \text{Normr3h}_{i,t} + \beta_3 \text{NormFluctMonth}_{i,t} \\ & + \beta_4 \text{SchType}_i + \beta_5 \text{SchLocation}_i + \beta_6 \text{Gender}_i \\ & + \sum_{g=1}^5 \rho_g G_{i,g} + \beta_{\text{IMR}} \text{IMR}_i \\ & + \text{DistrictFE}_i + \text{DateFE}_t + \varepsilon_{i,t}. \end{aligned} \quad (8)$$

Moderation model. To assess heterogeneous effects of climate by school type and location, we interact each climate variable in $\mathcal{T}$ with each moderator in $\mathcal{M}$, where

$$\mathcal{T} = \{\text{Normr1h}, \text{Normr3h}, \text{NormFluctMonth}\}, \quad \mathcal{M} = \{\text{SchType}, \text{SchLocation}, \text{Gender}\}. \quad (9)$$

The moderation model is

$$\begin{aligned} Y_{i,t} = & \beta_0 + \beta_1 \text{Normr1h}_{i,t} + \beta_2 \text{Normr3h}_{i,t} + \beta_3 \text{NormFluctMonth}_{i,t} \\ & + \beta_4 \text{SchType}_i + \beta_5 \text{SchLocation}_i + \beta_6 \text{Gender}_i \\ & + \sum_{T \in \mathcal{T}} \sum_{M \in \mathcal{M}} \theta_{TM}(T_{i,t} M_i) \\ & + \sum_{g=1}^5 \rho_g G_{i,g} + \beta_{\text{IMR}} \text{IMR}_i \\ & + \text{DistrictFE}_i + \text{DateFE}_t + \varepsilon_{i,t}. \end{aligned} \quad (10)$$

Linear double machine learning. To guard against functional-form misspecification in the high-dimensional fixed-effect environment, we re-estimate equation (10) as a LinearDML model^{21,32}. Here, Y denotes either of the two educational outcomes, `PassRate` or `ExcellenceRate`, and T denotes one focal climate indicator at a time, namely `Normr1h`, `Normr3h` or `NormFluctMonth`. In Stage 1, LassoCV regresses Y and each T on the full set of controls $\mathbf{W}$ (the Inverse Mills Ratio, the other two climate indicators, grade dummies, district dummies and school-year dummies) and computes residuals $\tilde{Y} = Y - \hat{f}(\mathbf{W})$ and $\tilde{T} = T - \hat{g}(\mathbf{W})$. In Stage 2, the treatment effect is estimated as a linear function of school type, school location and gender:

$$\tilde{Y}_{i,t} = (\theta_0 + \theta_1 \text{SchType}_i + \theta_2 \text{SchLocation}_i + \theta_3 \text{Gender}_i) \tilde{T}_{i,t} + \eta_{i,t}. \quad (11)$$

This procedure satisfies the Neyman orthogonality condition and yields $\sqrt{n}$ -consistent, asymptotically normal coefficient estimates even when $\hat{f}$ and $\hat{g}$ are estimated by regularised, biased learners^{21,33}.

Causal forest. To recover heterogeneous treatment effects without a parametric interaction form, we train a CausalForestDML^{22,34} for each of the six outcome $\times$ treatment combinations.

For a continuous standardised rainfall treatment T , the target is the marginal conditional average treatment effect

$$\tau(x) = \frac{\partial E[Y_i \mid do(T_i = t), X_i = x]}{\partial t}, \quad (12)$$

where $X_i = (\text{SchType}_i, \text{SchLocation}_i, \text{Gender}_i)$. As in the source causal-forest specification, the model first residualises outcome and treatment,

$$\tilde{Y}_i = Y_i - \hat{m}(X_i, W_i), \quad \tilde{T}_i = T_i - \hat{e}(X_i, W_i), \quad (13)$$

and then estimates the orthogonal relation $\tilde{Y}_i = \tau(X_i)\tilde{T}_i + \varepsilon_i$. For any target group x , honest trees induce adaptive weights $\alpha_i(x)$, giving the local weighted residual regression

$$\hat{\tau}(x) = \frac{\sum_i \alpha_i(x) \tilde{T}_i \tilde{Y}_i}{\sum_i \alpha_i(x) \tilde{T}_i^2}. \quad (14)$$

The nuisance models use LassoCV and the control vector W contains the Inverse Mills Ratio, the other two rainfall indicators, grade dummies, district dummies and date dummies. We use 2,000 honest trees, minimum leaf size 100, a minimum within-leaf treatment-variance fraction of 0.1 and three-fold cross-fitting. We evaluate $\hat{\tau}(x)$ at each of the eight combinations of (SchType, SchLocation, Gender) to obtain group-level CATEs (Fig. 4), and extract normalised feature-importance scores from the trained forest to quantify which moderator drives most of the heterogeneity (Fig. 5).

Robustness checks

We assess robustness on three dimensions. (i) Winsorisation: we replace observations of Normr1h, Normr3h and NormFluctMonth above the 99th or below the 1st percentile (and analogously at 5%/95%) with the threshold value and re-estimate. (ii) Effective sample: we restrict to school-round observations with more than 10 (or 15) dictation participants to limit small-sample noise in the rates. (iii) Finer-grained fixed effects: we replace District fixed effects with the finer Region fixed effects.

Software and reproducibility

LLM evidence mining used the DeepSeek-V3.2 API. Data processing used Python 3.12 with `pandas`, `numpy` and `ocrmypdf`. Fixed-effects regressions used Stata 19 (`reghdfe` and `xtreg`). LinearDML and CausalForest models used Python’s EconML package (v0.16) with `scikit-learn` LassoCV and `statsmodels`. Random seeds were set to 120 for LinearDML and 32 for the causal forests. Re-running each step from the raw input recovers the reported numbers to within Monte-Carlo error.

Data and code availability

The aggregated and de-identified dataset and the analysis code are available at <https://github.com/Andy-611/climate-education-civ>. The underlying Daily Dictation micro-data are not pub-

licly redistributable because they are owned by UNICEF Côte d’Ivoire and MENA; requests for access can be submitted through UNICEF Côte d’Ivoire subject to formal data-sharing agreement. The CHIRPS-derived rainfall data are publicly available on the OCHA Humanitarian Data Exchange^{24,60}.

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Author contributions

A.Y. designed the study, accessed the data under institutional authorisation, built the LLM evidence-mining pipeline, performed the econometric and causal-forest analyses and wrote the first draft. Z.Z. supervised the research, advised on identification strategy and revised the manuscript. Both authors approved the final version.

Competing interests

The authors declare no competing interests.

Extended Data

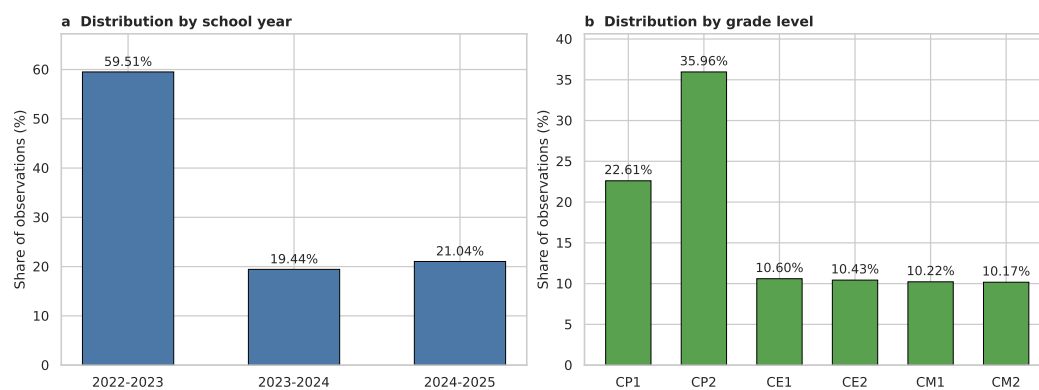


Figure ED1. Dataset composition by school year and grade level. Panel **a** shows the share of observations across the three covered school years (2022–2023, 2023–2024 and 2024–2025); the 2022–2023 share is larger because the Daily Dictation system was rolled out aggressively in its first year. Panel **b** shows the share by grade: pupils in the CP1 and CP2 grades dominate, consistent with the policy emphasis of the programme on early-grade literacy.

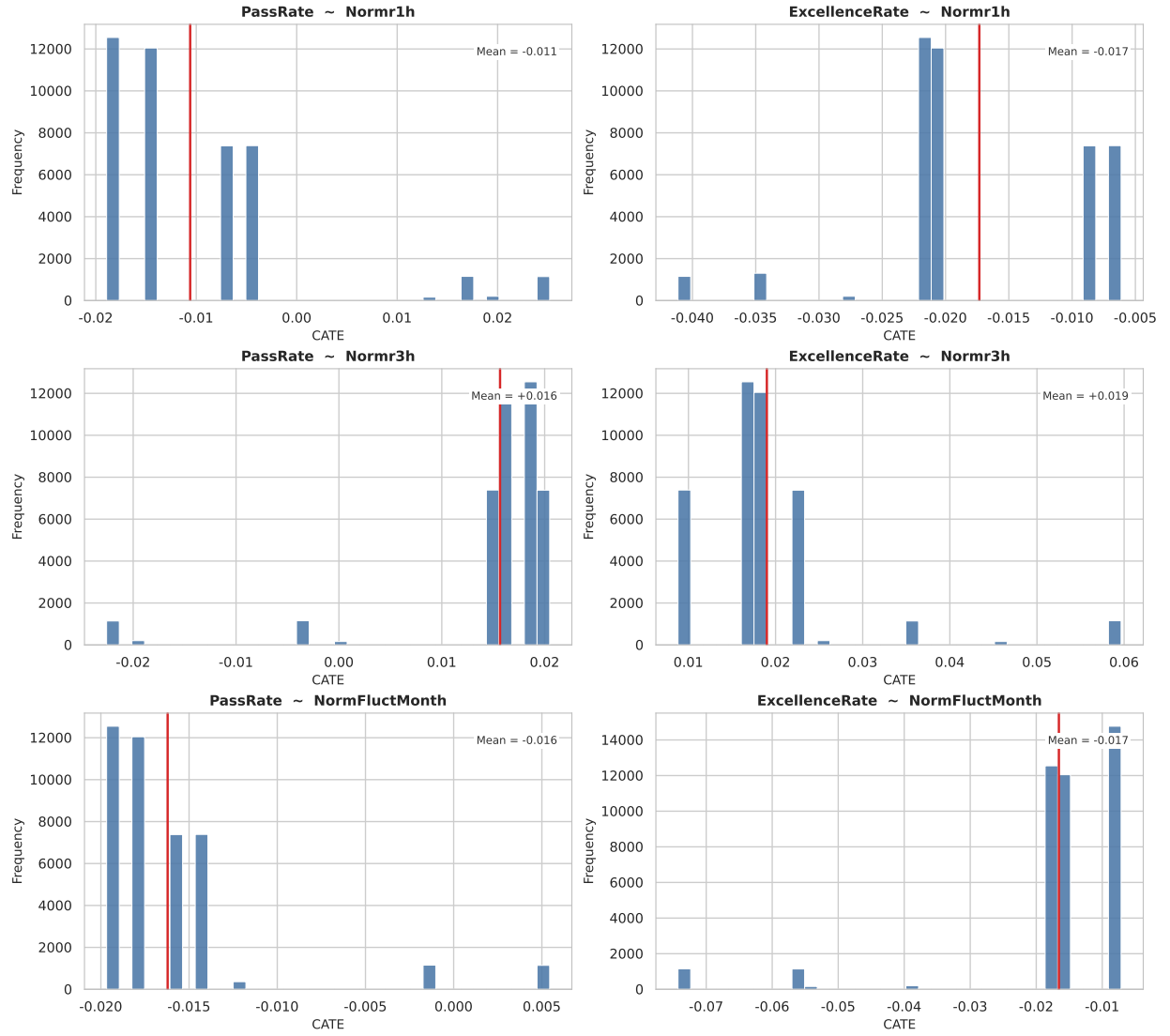


Figure ED2. Frequency distribution of conditional average treatment effects (CATEs) from the causal forests. For each of the six causal-forest models, the histogram shows the per-observation CATE distribution implied by the three discrete effect modifiers: school type, school location and student gender. Each bar corresponds to one subgroup CATE, the red line marks the sample-weighted mean. The distribution is multi-modal in every panel, indicating substantive heterogeneity rather than mass at zero.

Table ED1. Descriptive statistics of the main analytical variables.

Variable	N	Mean	SD	Min	Q1	Median	Q3	Max
PassRate (%)	64,743	0.758	0.245	0.000	0.611	0.817	0.996	1.000
ExcellenceRate (%)	64,743	0.402	0.287	0.000	0.153	0.391	0.625	1.000
r1h (mm)	64,743	61.40	64.67	2.18	12.69	35.48	87.60	259.27
r3h (mm)	64,743	266.90	171.64	10.23	103.65	272.23	400.91	713.18
FluctMonth	64,743	2.942	7.162	0.000	0.534	0.741	2.384	49.66
Gender (male=1)	64,743	0.500	0.500	0	0	0	1	1
SchType (private=1)	64,743	0.059	0.236	0	0	0	0	1
SchLocation (urban=1)	64,743	0.353	0.478	0	0	0	1	1
IMR (in-programme)	64,743	0.640	0.599	0.030	0.143	0.422	0.976	3.060

Notes. Sample is the full Daily Dictation data after matching with sub-national rainfall, October 2022 to June 2025. The Inverse Mills Ratio (IMR) is computed from the first-stage Heckman selection model and captures the predicted non-randomness of school participation and successful reporting in the Daily Dictation programme.

Table ED2. Pearson correlation matrix of the main analytical variables.

	Normr1h	Normr3h	NormFluctMonth	SchType	SchLocation	Gender	IMR
Normr1h	1.000						
Normr3h	0.494	1.000					
NormFluctMonth	-0.248	0.166	1.000				
SchType	0.107	0.079	-0.045	1.000			
SchLocation	0.019	0.028	-0.035	0.251	1.000		
Gender	0.001	0.001	0.000	0.001	0.000	1.000	
IMR	0.006	0.068	0.016	0.271	0.080	0.001	1.000

Table ED3. Variance-inflation factors (VIFs) of the main analytical variables.

Variable	VIF
Normr1h	1.55
Normr3h	1.46
NormFluctMonth	1.26
SchType	1.17
SchLocation	1.07
Gender	1.00
IMR	1.10

Table ED4. Heckman first-stage selection equation: who joins and reports?

Variable	Probit on $D_i = 1$
SchLocation (urban=1)	0.138*** (0.000)
SchType (private=1)	-0.855*** (0.000)
Electricity	0.157*** (0.000)
Multimedia room	0.156 (0.136)
Internet connection	0.489*** (0.000)
District participation density	3.940*** (0.000)
Constant	-1.957*** (0.000)
N (schools)	19,495

Notes. Robust-SE-based p -values in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table ED5. Main-effect estimates across pooled OLS, two-way fixed-effects, and Heckman-corrected specifications.

Variable	Pooled OLS		Two-way FE		Two-way FE + IMR	
	PassRate	ExcellenceRate	PassRate	ExcellenceRate	PassRate	ExcellenceRate
Normr1h	−0.012***	−0.009***	−0.005*	−0.016***	−0.005**	−0.016***
Normr3h	−0.006***	−0.012***	0.019***	0.028***	0.020***	0.029***
NormFluctMonth	−0.024***	−0.023***	−0.017***	−0.014***	−0.018***	−0.015***
SchType (private=1)	0.073***	0.064***	0.058***	0.037***	0.066***	0.042***
SchLocation (urban=1)	0.000	0.034***	−0.009***	0.020***	−0.010***	0.019***
Gender (male=1)	−0.011***	−0.009***	−0.011***	−0.009***	−0.011***	−0.009***
IMR	−	−	−	−	−0.017***	−0.009***
District FE	No	No	Yes	Yes	Yes	Yes
Date FE	No	No	Yes	Yes	Yes	Yes
N	64,743	64,743	64,743	64,743	64,743	64,743
R^2	0.047	0.147	0.081	0.186	0.082	0.187
Adj. R^2			0.022	0.079	0.023	0.079

Notes. Robust-SE-based p -values used to mark significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Grade dummies and the constant are omitted.

Table ED6. Moderation by household economic conditions and gender (interaction model).

	PassRate	ExcellenceRate
Normr1h	−0.011***	−0.018***
Normr3h	0.025***	0.031***
NormFluctMonth	−0.024***	−0.017***
SchType (private=1)	0.064***	0.040***
SchLocation (urban=1)	−0.009***	0.019***
Gender (male=1)	−0.011***	−0.009***
IMR	−0.017***	−0.010***
Normr1h × SchType	0.007	−0.017***
Normr1h × SchLocation	0.014***	0.009***
Normr1h × Gender	−0.001	−0.001
Normr3h × SchType	−0.001	0.016***
Normr3h × SchLocation	−0.009***	−0.011***
Normr3h × Gender	0.000	0.001
NormFluctMonth × SchType	0.005	−0.026***
NormFluctMonth × SchLocation	0.016***	0.008***
NormFluctMonth × Gender	0.001	0.001
District FE / Date FE	Yes / Yes	Yes / Yes
N	64,743	64,743
R^2	0.083	0.187

Notes. Robust-SE-based p -values used to mark significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Grade dummies are included but suppressed.

Table ED7. LinearDML estimates of the moderation effects. Cross-fitted (3 folds) with LassoCV nuisance estimators.

	PassRate	ExcellenceRate
$\tilde{\text{Normr1h}} \times \text{SchType}$	0.007	-0.026***
$\tilde{\text{Normr1h}} \times \text{SchLocation}$	0.006*	0.000
$\tilde{\text{Normr1h}} \times \text{Gender}$	0.000	0.000
$\tilde{\text{Normr3h}} \times \text{SchType}$	0.004	0.021***
$\tilde{\text{Normr3h}} \times \text{SchLocation}$	-0.006**	-0.008***
$\tilde{\text{Normr3h}} \times \text{Gender}$	0.001	0.001
$\tilde{\text{NormFluctMonth}} \times \text{SchType}$	0.005	-0.038***
$\tilde{\text{NormFluctMonth}} \times \text{SchLocation}$	0.007**	0.005
$\tilde{\text{NormFluctMonth}} \times \text{Gender}$	0.000	0.002
N	64,743	64,743

Notes. Tilded variables denote LassoCV-residualised quantities. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.